Linux Task Documentation – File Ownership & Permissions

Date: 9 February 2026

Objective:

The objective of this task was to understand Linux file ownership, group ownership, permission errors, and how to resolve them using sudo privileges.

Environment:

- OS: Linux (Ubuntu / WSL)

- Terminal: Bash

- Users involved: bob, shariqua

- Group involved: testgroup

Task Summary:

- Created a file using a normal user

- Faced permission denied error while changing ownership

- Used sudo to change file owner and group

- Verified file ownership and permissions

- Applied secure permission settings

Steps Performed:

1. Created a File:

touch report.txt

2. Checked File Details:

ls -l

Output:

-rw-rw-r-- 1 bob bob report.txt

3. Attempted to Change Ownership (Failed):

chown shariqua report.txt

Error:

Operation not permitted

4. Changed Ownership Using sudo:

sudo chown shariqua report.txt

5. Verified Ownership:

ls -l

Output:

-rw-rw-r-- 1 shariqua bob report.txt

6. Changed Group Ownership:

sudo chown :testgroup report.txt

7. Verified Group Change:

ls -l

Output:

-rw-rw-r-- 1 shariqua testgroup report.txt

8. Changed File Permissions:

sudo chmod 640 report.txt

9. Final Permission Check:

ls -l

Final Output:

-rw-r----- 1 shariqua testgroup report.txt

Permission Explanation:

6 = Read + Write (Owner)

4 = Read only (Group)

0 = No access (Others)

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bob@DESKTOP-8UVUP6Q: ~
bob@DESKTOP-8UVUP6Q:/home/shariqua$ cd
bob@DESKTOP-8UVUP6Q:~$ touch report.txt
bob@DESKTOP-8UVUP6Q:~$ ls -l
total 0
-rw-rw-r-- 1 bob bob 0 Feb  9 01:30 report.txt
bob@DESKTOP-8UVUP6Q:~$ chown shariqua report.txt
chown: changing ownership of 'report.txt': Operation not permitted
bob@DESKTOP-8UVUP6Q:~$ sudo chown shariqua report.txt
[sudo] password for bob:
bob@DESKTOP-8UVUP6Q:~$ su shariqua
Password:
shariqua@DESKTOP-8UVUP6Q:/home/bob$ cd
shariqua@DESKTOP-8UVUP6Q:~$ ls -l
total 0
shariqua@DESKTOP-8UVUP6Q:~$ su bob
Password:
bob@DESKTOP-8UVUP6Q:/home/shariqua$ cd
bob@DESKTOP-8UVUP6Q:~$ sudo chown shariqua report.txt
bob@DESKTOP-8UVUP6Q:~$ ls -l
total 0
-rw-rw-r-- 1 shariqua bob 0 Feb  9 01:30 report.txt
bob@DESKTOP-8UVUP6Q:~$ sudo chown :testgroup report.txt
bob@DESKTOP-8UVUP6Q:~$ ls -l
total 0
-rw-rw-r-- 1 shariqua testgroup 0 Feb  9 01:30 report.txt
bob@DESKTOP-8UVUP6Q:~$ sudo chmod 640 report.txt
bob@DESKTOP-8UVUP6Q:~$ ls -l
total 0
-rw-r----- 1 shariqua testgroup 0 Feb  9 01:30 report.txt
bob@DESKTOP-8UVUP6Q:~$
```

Conclusion:

This task demonstrates practical handling of Linux file ownership, group management, permission errors, and secure access control using sudo.

Author:

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